

Monograph Summary

This monograph provides the foundational mathematical theory and numerical algorithms for nonlinear multiobjective optimization. It formally defines Pareto optimality, non-dominated sets, trade-off analysis, scalarization techniques (weighting methods, epsilon-constraint methods, achievement scalarizing functions), and interactive decision-making algorithms under conflicting criteria.

Role in Our VLM-DB Multi-Vector Storage Design (RQ2)

Used in Section 4.2 to mathematically formalize Pareto optimal configurations across the 4-dimensional objective space (retrieval quality nDCG@10, P95 latency, storage footprint, and ingestion cost), deriving the 28 non-dominated configurations out of 112 compatible system designs.